

Infosys Springboard Virtual Internship 7.0

Completion Report

Team Details <Do not mention any personally identifiable information like email ID, institute details, mobile phone number etc.>

Batch Number

Start date

Names: Thota Sai Venkat

Internship Duration: 8 Weeks

1. Project Title

Infosys _Healthcare Management Platform for Clinical Operations.

2. Project Objective

The key goal of Medisphere project is to develop an advanced full-stack healthcare platform which helps in making appointment scheduling easier for patients and doctors and gives administrators full control on system management. Some of the main aspects include onboarding of patients, appointment management, creation of digital receipt for payments, inclusion of AI functionalities and analysis of system logs.

3. Project description in detail

The project was built on the basis of a decoupled architecture with a dynamic React front-end for smooth interactions and scalable Java back-end. Data consistency and relational mapping were defined through the use of a well-defined database schema.

Tools Used:

Front-end: React, HTML5, CSS3, JavaScript

Back-end: Java, Core Spring Framework

Database: MySQL

AI & Utilities: Artificial Intelligence Model integration, REST APIs,, Git

Impact of Application: Medisphere decreases patient waiting time, makes scheduling of doctors easier, provides digital receipt of payments, and utilizes artificial intelligence capabilities.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Requirement analysis, project setup, and UI wireframing.	Defined project requirements and designed user interface layouts for

		login/signup portals.
Week 2	Frontend UI development for core user roles.	Created UI for Admin login, Patient login/signup, and doctor selection for booking appointments.
Week 3	Database schema architecture and relational modeling.	Designed and deployed complete database schema covering Admins, Doctors, Patients, and Bookings.
Week 4	Backend API development and Spring framework integration.	Developed core backend RESTful services mapped to the database architecture.
Week 5	Feature integration: Payments and Doctor management.	Implemented payment receipt download feature and doctor appointment acceptance functionality.
Week 6	Administrative dashboard and system log analysis.	Developed Admin management features and integrated system log analysis for monitoring.
Week 7	AI feature development and integration.	Integrated live AI functionality into the application and verified full operational execution.
Week 8	End-to-end testing, documentation, and final report.	Conducted system testing, finalized project documentation, and prepared submission report.

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Finalized scope, tools, and UI wireframes for Medisphere.	July 08, 2026
Prototype/First Draft	Developed Admin and Patient login/signup portals and appointment booking interface.	July 08, 2026
Mid-Term Review	Designed database architecture and integrated backend APIs with React for payments and doctor portal.	July 21, 2026
Final Submission		
Presentation		

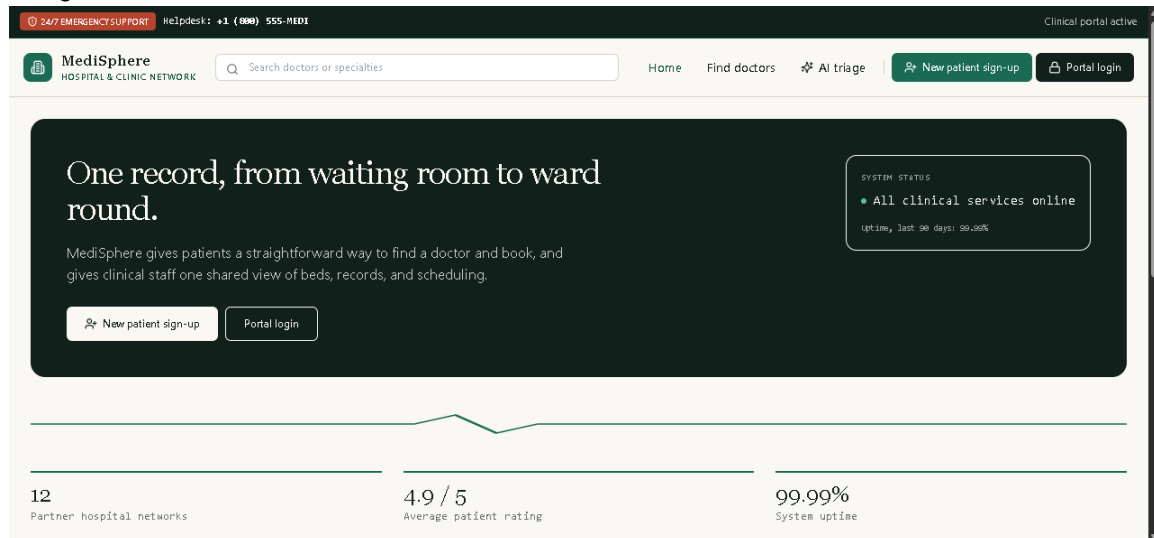
5b. Project execution details

The project followed an Agile development lifecycle divided into clear iterative sprints across 8 weeks:

- **Architecture & Database Setup:** Established a decoupled application architecture. Standardized backend entities (Admins, Doctors, Patients, Bookings) using MySQL relational mapping to handle scalable database transactions.
- **Frontend Development:** Built a responsive single-page frontend using React, custom CSS styling, and component modularity. Specialized views were created for Public Landing, Doctor Management, and Patient Portals.
- **Backend API Layer:** Implemented RESTful APIs leveraging Core Spring Framework. Integrated controllers, repositories, and data handling workflows for appointment booking, status updates, and dynamic bill generation.
- **AI & Administrative Features:** Integrated an AI model for intelligent triage support. Developed system log tracking and custom reporting panels within the Admin dashboard for system-wide auditing.
- **Verification & Testing:** Performed end-to-end integration testing across frontend components and backend REST API endpoints to confirm complete data flow stability.

6. Snapshots / Screenshots

Include relevant visuals such as screenshots of work done, dashboards, code snippets, or designs.



24/7 EMERGENCY SUPPORT

helpdesk: +1 (800) 555-MEDI

Clinical portal active

MediSphere

HOSPITAL & CLINIC NETWORK

[Home](#) [Find doctors](#) [AI triage](#) [New patient sign-up](#) [Portal login](#)

AI Clinical Triage Assistant

Routing suggestions only, not a medical diagnosis.

Run AI Triage Analysis

Possible emergency: If symptoms are severe, call emergency services immediately.

Suggested department: Cardiology

Matched symptoms: chest pain

Suggested specialist: Dr. Sai Venkat (Cardiology)

Request an appointment with Dr. Sai Venkat

This is an automated triage-routing suggestion, not a medical diagnosis. For emergencies, contact your local emergency number immediately.

Patient Portal Dashboard

Welcome back, Gayathri

Join Telehealth Call

Sign Out

Appointments & Booking

Previous Bookings (2)

Lab Reports (1)

Invoices & Checkout

Health Records (EHR)

Prescriptions & Refills

Book Specialist Consultation

Select your preferred physician, appointment date, and available time slot.

Select Doctor & Specialty

Dr. Nagalakshmi (General Practice) — \$249

Consultation Date

26-08-2026

Available Time Slots

09:00 AM

09:30 AM

10:00 AM

10:30 AM

11:00 AM

11:30 AM

12:00 PM

12:30 PM

01:00 PM

01:30 PM

02:00 PM

02:30 PM

03:00 PM

03:30 PM

04:00 PM

04:30 PM

Reason for Visit / Symptoms Summary

Describe symptoms (e.g. persistent cough, routine cardiology follow-up)...

Confirm & Book Appointment

Scheduled Upcoming Appointments

Admin & Hospital Operations

System Admin • Full System Access

Sign Out

ADMIN WORKSPACE

Dashboard Overview

Register Patient

Onboard Doctor

Patient Directory

Doctor Roster

Pharmacy & Supplies

Lab & Diagnostics

Billing & Invoices

Admin Access

System Logs

Hospital Analytics & Executive Dashboard

Live operational overview, financial KPIs, and facility capacity metrics.

TOTAL REVENUE

\$

\$4,896

\$3,298 collected

+8.4%

ACTIVE APPOINTMENTS

5

4 completed

+12%

PATIENT CENSUS

3

0 Inpatients

PHARMACY ALERTS

0

0 out of stock

Weekly Booking & Revenue Trends

Live 7-day revenue performance and patient volume

Revenue (\$)

Appointments

Thu

Fri

Sat

Sun

Mon

Tue

Wed

Patient Census & Acuity

Inpatient vs Outpatient admission ratio & condition acuity

3 PATIENTS

Inpatient (Admitted)

3 patients (100%)

Outpatient

0 patients (0%)

Stable

0

Recd

0

Crit

0

pg. 4


Doctor Clinical Portal
 Dr. Ashok Rajgopal • Orthopedics

Sign Out

Triage Queue (0)
 Patient Registry
 E-Prescription Writer
Lab & Diagnostics (0)
 Schedule & Availability
 Practice Analytics

Issue Certified E-Prescription
Connects directly to live pharmacy inventory. Quantities will be deducted upon pharmacy dispensing.

Select Patient

Sai Venkat Thota (Not Specified, 22y) — Not Specified

Select Medication from Pharmacy Inventory

-- Choose from live inventory or type below --

Medication Name	Prescription Quantity
e.g., Amoxicillin 500mg	1

Dosage & Instructions

Standard Dose

Sign & Transmit E-Prescription

7. Challenges Faced

• Managing State Across Decoupled Architecture:

- *Challenge:* Synchronizing UI state updates dynamically when doctors update appointment statuses or room assignments.
- *Resolution:* Implemented centralized state management contexts in React to handle asynchronous API response updates gracefully.

• AI Model API Integration:

- *Challenge:* Structuring real-time responses from the AI integration without blocking primary application workflows.
- *Resolution:* Applied async REST handling to parse model responses smoothly without UI degradation.

• Complex Data Formatting for Digital Receipts:

- *Challenge:* Formatting dynamic patient billing data into structured, downloadable receipts across diverse patient interactions.
- *Resolution:* Developed specialized billing helper utilities to format transactional attributes consistently.

8. Learnings & Skills Acquired

- **Technical Technologies:** Full-stack development utilizing Java and Core Spring Framework for back-end APIs, alongside modern React, JavaScript, HTML5, and CSS3 for interactive interfaces.

- **Database & Architecture Management:** Relational database design in MySQL, RESTful web service architecture, and Git version control workflows.
- **Domain Knowledge:** Operational healthcare IT processes including IPD/OPD patient workflows, digital billing, appointment triage, and clinical scheduling rules.
- **Professional Skills:** Project execution planning, system testing, software documentation, and analytical problem-solving.

9. Testimonials from team

Working on Medisphere during the Infosys Springboard Virtual Internship provided hands-on experience in building a real-world enterprise application. Translating clinical workflows into code and integrating intelligent features like AI triage was an incredible technical journey that strengthened my confidence in full-stack Java development.

10. Conclusion

The Infosys Springboard Virtual Internship 7.0 provided an exceptional platform to build **Medisphere**, a robust full-stack healthcare management application. By seamlessly bridging core backend Java architecture with modern React interfaces, the project addresses key bottlenecks in hospital management such as patient scheduling and administrative log monitoring. This experience significantly bolstered full-stack software development competencies and directly aligned with career goals in backend engineering and complex system integration.

11. Acknowledgements

I express my sincere gratitude to **Infosys Springboard** for offering this virtual internship opportunity. Special thanks to our project mentors and advisors for their guidance, domain insights, and continuous feedback throughout the 8-week journey. I also appreciate the collaborative efforts of fellow interns and team members who supported the successful completion of this project.